Candidate name:

_____

Date: _____

Unofficial NUFYPET Math Simulation

Time: 1 hour (60 min.) Version 6.1
Total Marks: ____/30

Instructions

- Circle the answer you find right
- Calculators **MUST NOT** be used
- Figures are **NOT** accurately drawn and **NOT** to scale, except it is mentioned in a question
- There is only one answer for each question, so answers with two or more selected options marked as zero
- If you think something is wrong with the test, report a Methodist

Information

- The total mark for test is 30 (Each question marked as 1 point)
- There is **NO** penalty for wrong answers

Advice

- Read each question carefully
- Follow the time (There is an average 2 minutes for each question)
- Don't stay in one task for long
- Try to answer for all questions
- Recheck your answers if possible

Note: This is unofficial test made by MSDC

1. The line L has equation $y = 2x - 1$.

Four of the following five points are the same distance from the line L. Which one is at a different distance?

A (1, -1) **B** (1, 3) **C** (4, 9) **D** (5, 13) **E** (6, 9)

2. The volume of water expands by 9% when it freezes.

If you want a block of ice $40\ cm^3$, how much water would you need to put into a freezer?

A $36.7\ cm^3$ **B** $40.0\ cm^3$ **C** $38.2\ cm^3$ **D** $38.0\ cm^3$ **E** $37.0\ cm^3$

3. What is the set of values of x for which $x^2 < 9$ and $2x + 3 \geq 5$

A $x > 3$ **B** $x > -3$ **C** $x > -1$ **D** $x < -3$ or $x \geq 1$ **E** $1 \leq x < 3$

4. A safe has external dimensions as follows:

Width $48cm$

Depth $44cm$

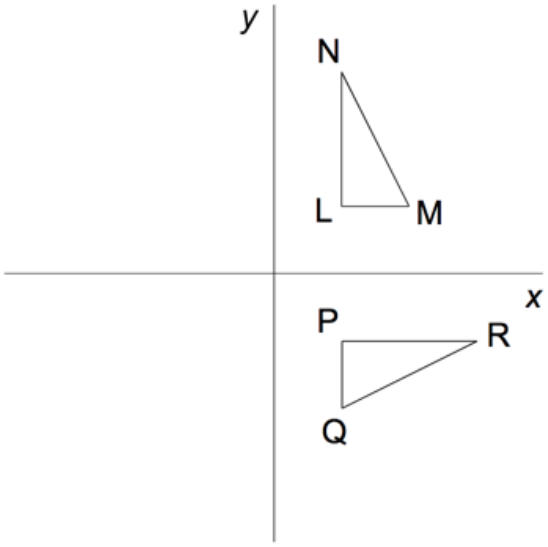
Height $52cm$

The entire safe is made of steel $4\ cm$ thick except the base which is $8\ cm$ thick.

What are the internal dimensions of the base of the safe?

A $40cm \times 40cm$ **B** $36cm \times 36cm$ **C** $40cm \times 32cm$ **D** $44cm \times 40cm$ **E** $40cm \times 36cm$

5. In the diagram below, the triangles are congruent, PQ is a continuation of NL and both L and P are equidistant from the line $x = 0$ and from the line $y = 0$.



Of the transformations (1 to 5) described below, which one of the combinations (A to E) will map the triangle LMN onto the triangle PQR?

1 Reflection in the line $y = 0$

2 Rotation of $90º$ anticlockwise about the origin

3 Reflection in the line $y = x$

4 Reflection in the line $x = 0$

5 Rotation of $180º$ anticlockwise about the origin

A 1 followed by 2 **B** 3 followed by 4 **C** 5 followed by 4 **D** 2 followed by 5 **E** 4 followed by 5

6. Simplify

$$\left(\frac{2x^{\frac{3}{2}}y^3}{\sqrt{z}}\right)^2$$

A $\frac{4x^6y^5}{z^2}$ **B** $\frac{4x^3y^6}{z}$ **C** $\frac{4x^3y^5}{z}$ **D** $\frac{4x^{\frac{5}{2}}y^5}{z}$

7 Find the positive solution of the following simultaneous equations:

$$4x^2 + y^2 + 10y = 47$$

$$2x - y = 5$$

A $x = 3\ y = 1$ **B** $x = 2\ y = 3$ **C** $x = 3\ y = 2$ **D** $x = 3\ y = 3$ **E** $x = 3\ y = 3$

8 a is inversely proportional to the square of b.

When $a = 9, b = 4$.

What is the positive value of b when $= 4$?

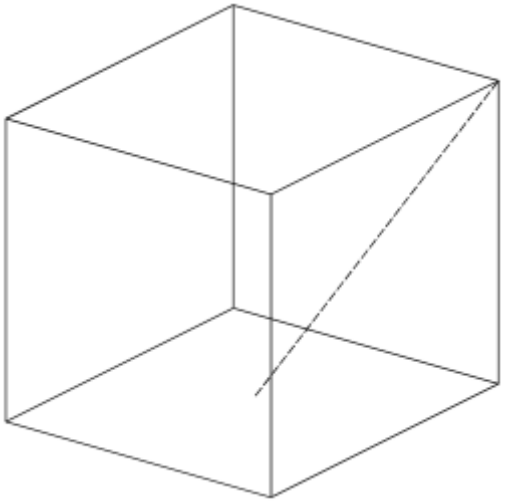
A 6 **B** 9 **C** 20.25 **D** 36

9 In a group of students, exactly ⅖ are male and exactly ⅓ study mathematics. The probability that a male student chosen at random from the group studies mathematics is p.

Which of the following is the range of possible values of p?

A $0 \leq p \leq {}^{1}/_{3}$ **B** ${}^{1}/_{3} \leq p \leq \frac{2}{5}$ **C** ${}^{1}/_{3} \leq p \leq 1$ **D** $0 \leq p \leq {}^{5}/_{6}$ **E** ${}^{2}/_{5} \leq p \leq {}^{5}/_{6}$

10. A cube has unit length sides. What is the length of a line joining a vertex to the midpoint of one of the opposite faces (the dashed line in the diagram below)?



A $\sqrt{2}$ **B** $\sqrt{\frac{3}{2}}$ **C** $\sqrt{\frac{5}{2}}$ **D** $\sqrt{3}$ **E** $\sqrt{5}$

11. Which of the following is the equation of the circle with centre $(1.5, 0.5)$ and radius 3?

A $2x^2 - 6x + 2y^2 + 2y - 13 = 0$

B $2x^2 + 6x + 2y^2 - 2y - 14 = 0$

C $2x^2 - 6x + 2y^2 - 2y - 13 = 0$

D $2x^2 + 6x + 2y^2 - 2y - 13 = 0$

E $2x^2 + 6x + 2y^2 - 2y - 23 = 0$

12. Solve the equation to find the value of x.

$$\frac{x+5}{2}-\frac{2x-1}{3}=4$$

A -5 **B** -6 **C** -7 **D** -8 **E** -9

13. What is the set of values for which $12-x^2>8$ and $2x+3\geq 5$?

A $1\leq x$ **B** $1<x\leq 2$ **C** $1\leq x<2$ **D** $2<x$ **E** $-1\leq x<2$

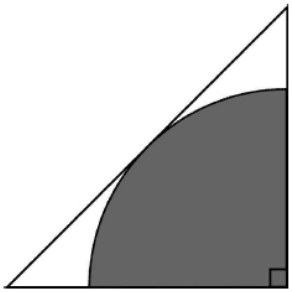
14. Calculate

$$\frac{3.246\times 10^6+3.246\times 10^4}{1.623\times 10^{11}}$$

Give your answer as an ordinary number, correct to 3 significant figures.

A 0.0000102 **B** 0.0000202 **C** 0.0000302 **D** 0.0000402 **E** 0.0000502

15. The diagram shows a quarter of a circle surrounded by an isosceles triangle.



The radius of the circle is r.

Which one of the following expressions represents the unshaded area?

A $\left(1-\frac{\pi}{4}\right)r^2$ **B** $\left(r^2-\frac{\pi r^2}{2}\right)$ **C** $\left(r^2-\frac{\pi r}{2}\right)$ **D** $(4-\pi)r^2$ **E** $\left(2-\frac{\pi}{4}\right)r^2$

16. Evaluate:

$$\sqrt{\frac{2\times 10^3+8\times 10^2}{\frac{1}{2500}+3\times 10^{-4}}}$$

A 20 **B** 140 **C** 200 **D** 625 **E** 2000

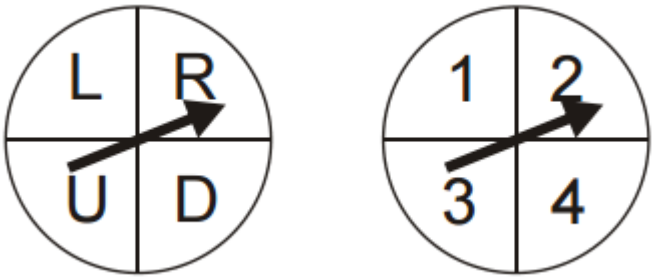
17. Rearrange $\sqrt{\frac{2}{x}+1}=y-3$ to make x the subject.

A $x=\frac{2}{(y-4)(y-2)}$ **B** $x=\frac{y-4}{2}$ **C** $x=\frac{(y-3)^2}{2}$ **D** $x=\frac{2}{(y+4)(y+2)}$

18. If $(x^2 - 16)/(x^2 + 6x + 8) = y$ and $x > -2$, which of the following is an expression for x in terms of y?

A $\frac{1+y}{2-y}$ **B** $\frac{2y+4}{1-y}$ **C** $\frac{4y-4}{y+1}$ **D** $\frac{2y-4}{2+y}$ **E** $\frac{y+4}{y+1}$

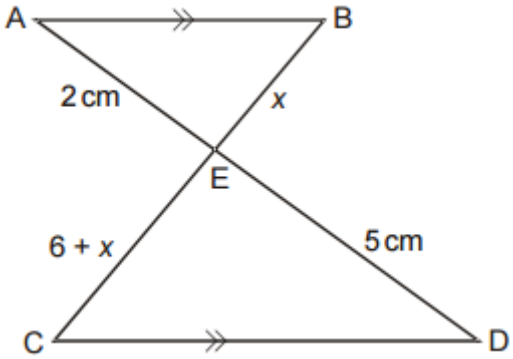
19. A children's game is played on a square grid starting in the centre. Players spin two spinners to decide how to move their counters. The first spinner decides the direction (Left, Right, Up or Down) and the second spinner decides the distance (1, 2, 3 or 4 squares).



What are the chances that, after two moves, a player is exactly back where they started?

A 1/256 **B** 1/64 **C** 1/16 **D** 1/8 **E** 1/4

20. AB is parallel to CD. (The diagram is not drawn to scale.)



Find the length of BE.

A $4cm$ **B** $8cm$ **C** $6cm$ **D** $10cm$ **E** $2cm$

21. The mean time for running a race by a group of 20 people was 54 seconds. The times for a second group of people were added and the value of the mean went up to 56 seconds.

Which formula represents the relationship between the number of people in the second group, P, and the mean time of the second group, T?

A $P = \frac{40}{T-54}$ **B** $P = \frac{1080}{(T-54)}$ **C** $P = \frac{40}{T-56}$ **D** $P = \frac{1080}{T-56}$ **E** $P = \frac{1120}{T} - 20$

22. The equation connects the variables A, x, y, z, P and Q.

$$A = \frac{(x+y)^2 z}{P} Q$$

The following changes are made:

x and y are both increased by $50\ \%$

z is decreased by $20\ \%$

P is doubled and Q remains the same.

What is the resulting percentage change in A?

A $10\ \%$ decrease **B** $10\ \%$ increase **C** $2.5\ \%$ increase **D** $2.5\ \%$ decrease **E** $20\ \%$ decrease

23. A circle has a diameter of $20cm$.

The line AC is a diameter of the circle.

B is a point on the circumference of the circle with $AB = 12cm$.

The point D lies on the arc AC on the opposite side to point B.

What is the sine of $\angle BDC$?

A 0.6 **B** 0.7 **C** 0.8 **D** 0.9 **E** 1.0

24. The oscillation period, t, for a spiral spring (weight w, radius R made from wire length l, radius r with a modulus of rigidity n) with a weight W hanging from it is given by:

$$t = 2\pi \sqrt{\frac{2lR^2\left(W + \frac{w}{3}\right)}{n\pi r^4 g}}$$

If such a spring is used to measure the weight, W, of an object we need to rearrange the equation so W is the subject.

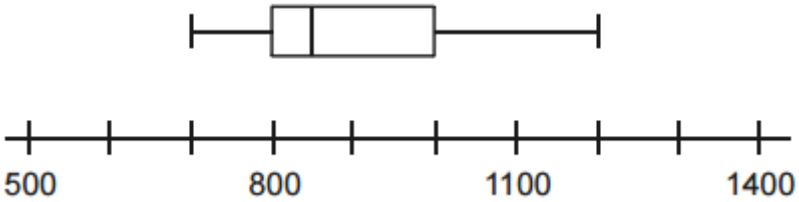
Which of the following is the correct rearrangement of the equation?

A $W = \frac{nr^4 gt^2}{8\pi lR^2} - \frac{w}{3}$ **B** $W = \frac{n\sqrt{n}r^4 gt^2}{8\pi lR^2} - \frac{w}{3}$ **C** $W = \frac{3nr^4 gt^2}{8\pi lR^2} - 3w$ **D** $W = \frac{n\sqrt{n}r^4 gt^2}{8\pi lR^2} - 3w$

25. If you look at a clock and the time is 9.45, what is the angle between the hour and the minute hands?

A 0° **B** 7.5° **C** 15° **D** 22.5° **E** 30°

26. The box and whisker plot shows some information about members of a health club using a walking machine. It shows the number of steps that a group of 30 members did in a period of 10 minutes.



Use this information to find the probability that when 3 of these 30 members are chosen at random, they will all have taken between 850 and 1000 steps.

A 3 / 4 **B** 150 / 100 **C** 1 / 8 **D** 3 / 30 **E** 1 / 64

27. In statistics Spearman's rank correlation coefficient is given by the formula:

$$r = 1 - \frac{6\sum d^2}{n(n^2-1)}$$

Rearrange the formula to make $\sum d^2$ the subject.

A $\sum d^2 = 1 - \frac{r(n^3-n)}{6}$

B $\sum d^2 = \left(\frac{(1-r)(n^3-n)}{6}\right)^2$

C $\sum d^2 = \frac{(1-r)(n^3-1)}{6}$

D $\sum d^2 = \frac{(1+r)(n^3-n)}{6}$

E $\sum d^2 = \frac{(1-r)(n^3-n)}{6}$

28. What is the equation of the straight line which passes through $(-6, 2)$ and is perpendicular to $4y + 3x = 8$?

A $3y - 4x = 18$ **B** $4y - 3x = 26$ **C** $3y + 4x = -18$ **D** $3y - 4x = 30$ **E** $3y + 4x = 30$

29. Evaluate:

$$\frac{8 \times 10^{-5}}{\sqrt{1.6 \times 10^7}} \times (1.2 \times 10^3)^2$$

A 2.88×10^{-3} **B** 2.88×10^{-2} **C** 2.88×10^{-5} **D** 2.88×10^{4} **E** 2.88×10^{-1}

30. Evaluate

$$\left(\frac{32^{\frac{1}{5}}+9^{0}}{81^{\frac{3}{4}}}\right)^{-1}$$

A 10 **B** 9 **C** 8 **D** 7 **E** 6

Answer key:

№	Correct Answer	Your answer
1	D	
2	A	
3	E	
4	E	
5	D	
6	B	
7	A	
8	A	
9	D	
10	B	
11	C	
12	C	
13	C	
14	B	
15	A	
16	E	
17	A	
18	B	
19	C	
20	A	
21	C	
22	A	
23	C	
24	A	
25	D	
26	E	
27	E	
28	D	
29	B	
30	B	
		Total: _____